Andhra University College of Engineering

Electrical and Electronics Engineering | Course Code: EE301

Power Systems Engineering - Laboratory Manual & Practical Guide

Syllabus Focus: EE301 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Transmission Line Parameters

" Syllabus Scope: Inductance and capacitance calculation for 1-phase and 3-phase lines, GMD and GMR formu

Unit 2: Performance of Transmission Lines

" Syllabus Scope: Short, Medium Nominal T and Pi, and Long lines, ABCD parameter derivation, Surge impeda

EE301: Power Systems Engineering - Advanced Topics & Assessment

Academic Track: Electrical and Electronics Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Symmetrical Fault Analysis

" Transients on transmission lines, Short circuit of synchronous machine, Selection of circuit breakers, Curre

Unit 4: Symmetrical Components & Unsymmetrical Faults

" Fortescue theorem, Sequence networks Positive, Negative, Zero, Single Line-to-Ground LG, Line-to-Line LL, an

Unit 5: Power Flow Analysis

" Bus classification Slack, PQ, PV, Formation of Y-bus matrix, Gauss-Seidel and Newton-Raphson power flow algo

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Determination of ABCD Parameters and Regulation of Medium Transmission Line Model

Lab 2: Measurement of Sequence Impedances of a 3-Phase Salient-Pole Synchronous Machine

Lab 3: Symmetrical and Unsymmetrical Fault Analysis on an AC Network Analyzer

Lab 4: Formation of Bus Admittance Matrix Y-bus and Newton-Raphson Load Flow Simulation

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* I.J. Nagrath, D.P. Kothari - Modern Power System Analysis McGraw-Hill 4th Ed
* Hadi Saadat - Power System Analysis PSA Publishing

Authorized NoteVault Educational Document " Verified for Registered University Students